Daily Logic Drill #1 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Statements & Quantifiers (∀/∃) · Negating Compound Statements · Negating Quantified Statements · Order of Quantifiers.

Section A Rapid Recognition

1. Negate: "For all integers n, $n^2 \geq 0$."

Answer: "There exists an integer n such that $n^2 < 0$."

Method: Negating $\forall n, P(n)$ gives $\exists n, \text{not } P(n)$; here $P(n)$ is "$n^2 \geq 0$" and its negation is "$n^2 < 0$".

Investigate further: This negation happens to be false (no integer squares to a negative value) — negation is a mechanical operation, independent of whether the result is true.

2. Negate: "There exists a real number x such that $x^2 = -1$."

Answer: "For all real numbers x, $x^2 \neq -1$."

Method: Negating $\exists x, P(x)$ gives $\forall x, \text{not } P(x)$; here not $P(x)$ is "$x^2 \neq -1$".

Investigate further: This negation is true: every real square is non-negative, so $x^2 = -1$ has no real solution.

3. Negate: "n is even and n is prime."

Answer: "n is odd, or n is not prime."

Method: De Morgan's law: not(P and Q) = (not P) or (not Q).

Investigate further: The connective flips from "and" to "or" — the single most common source of negation mistakes (see A6).

4. Negate: "n is a multiple of 4 or n is a multiple of 6."

Answer: "n is not a multiple of 4, and n is not a multiple of 6."

Method: De Morgan's law: not(P or Q) = (not P) and (not Q).

Investigate further: The mirror image of A3: "or" flips to "and."

5. True or false: "The negation of 'for all x, $P(x)$' is 'for all x, not $P(x)$'."

Answer: False.

Method: Negating a ∀ gives an ∃, not another ∀: the correct negation is "there exists x such that not $P(x)$".

> **Investigate further:** Keeping the same quantifier when negating is the single most common error with quantified statements — the quantifier itself must flip.

6. True or false: "not(P and Q) is the same as (not P) or (not Q)."

Answer: True.

Method: This is exactly De Morgan's law, matching A3's worked example.

> **Investigate further:** Section B, B8 extends this to a nested compound statement with three atoms.

7. Negate: "$x > 3$ and $x < 10$."

Answer: "$x \leq 3$ or $x \geq 10$."

Method: De Morgan's law applied to the compound inequality: negate each inequality (strict flips to non-strict, opposite direction) and flip "and" to "or".

> **Investigate further:** Forgetting the "or equal to" when negating a strict inequality is a common slip — $x < 3$ would wrongly drop $x = 3$.

8. Is the statement "some prime numbers are even" a $\forall$-statement or a $\exists$-statement in form?

Answer: $\exists$-statement.

Method: "Some" means "there exists at least one", so this reads as "there exists a prime number that is even" — an existential claim, and a true one, witnessed by 2.

> **Investigate further:** Recognising "some", "at least one", and "there is a" as all signalling $\exists$ is essential before Section B's nested statements.

9. True or false: "$\exists x\, \forall y : x + y = 0$" and "$\forall y\, \exists x : x + y = 0$" (both over the reals) mean the same thing.

Answer: False.

Method: $\exists x\, \forall y : x + y = 0$ claims a single x works for every y — false, since $x + y = 0$ would need to hold even as y varies freely. $\forall y\, \exists x : x + y = 0$ claims that for each y a (possibly different) x exists — true, taking $x = -y$ every time.

> **Investigate further:** Swapping the order of $\exists$ and $\forall$ can turn a false statement into a true one — this is the central theme of today's toolkit, developed fully in Section C's C1 and Section D's D5.

10. Negate: "n is prime or $n = 1$."

Answer: "n is not prime, and $n \neq 1$."

Method: De Morgan's law: not(P or Q) = (not P) and (not Q).

> **Investigate further:** This negation describes exactly the composite numbers greater than 1 — a useful way to read "not prime and not 1" concretely.

Section B Manipulation Drills

1. Consider the statement about Fred: "Every day next week, Fred will do at least one maths problem." State the negation.

Answer: "Some day next week, Fred will do no maths problems."

Method: Negating $\forall d, P(d)$ gives $\exists d, \text{not } P(d)$; here $P(d)$ is "Fred does at least one problem on day d", so not $P(d)$ is "Fred does no problems on day d".

> **Investigate further:** "At least one" is itself an $\exists$-flavoured clause nested inside the outer $\forall$ — negating the whole statement only flips the outer quantifier and the inner clause, not the day-by-day structure.

2. True or false: "not(P or Q or R) is the same as (not P) and (not Q) and (not R)."

Answer: True.

Method: De Morgan's law extends to any number of terms: negating a chain of "or"s gives a chain of "and"s of the individual negations.

> **Investigate further:** The three-term version used directly in B3.

3. Negate: "n is a multiple of 2 and n is a multiple of 3 and n is a multiple of 5."

Answer: "n is not a multiple of 2, or n is not a multiple of 3, or n is not a multiple of 5."

Method: Extended De Morgan: negating a chain of "and"s gives a chain of "or"s of the individual negations.

> **Investigate further:** At least one of the three inner clauses must fail — not necessarily all three.

4. Negate: "For every real $x > 0$, there exists a real $y > 0$ with $y < x$."

Answer: "There exists a real $x > 0$ such that for every real $y > 0$, $y \geq x$."

Method: Negating $\forall x{>}0\, \exists y{>}0, P(x, y)$ gives $\exists x{>}0\, \forall y{>}0, \text{not } P(x, y)$; here not $P(x, y)$ is "$y \geq x$".

> **Investigate further:** The original statement is true (halve x to get a smaller positive y), so this negation is correctly false — no positive x is a strict lower bound for all positive y.

5. True or false: "'$\forall n\, \exists m : m > n$' and '$\exists m\, \forall n : m > n$' (both over the integers) mean the same thing."

Answer: False.

Method: $\forall n\, \exists m : m > n$ is true: for every integer, a larger integer exists. $\exists m\, \forall n : m > n$ is false: no single integer m exceeds every integer n (take $n = m$ itself, or larger).

> **Investigate further:** The same swapped-quantifier trap as A9, now over an unbounded domain rather than an equation.

6. Negate the nested statement: "There exists a prime p such that for every prime $q > p$, q is odd." Is the original statement true or false?

Answer: Negation: "For every prime p, there exists a prime $q > p$ such that q is even." The original statement is true.

Method: Negating $\exists p\, \forall q{>}p, P(q)$ gives $\forall p\, \exists q{>}p, \text{not } P(q)$. The original is true: taking $p = 2$, every prime $q > 2$ is automatically odd, since 2 is the only even prime.

> **Investigate further:** 2's exceptional evenness is exactly what makes the original true — the negation (false) would require infinitely many even primes larger than some bound, which don't exist.

7. Negate: "n is a multiple of 3, or (n is a multiple of 2 and n is a multiple of 5)."

Answer: "n is not a multiple of 3, and (n is not a multiple of 2 or n is not a multiple of 5)."

Method: Outer De Morgan on the "or" gives "(not multiple of 3) and not(multiple of 2 and multiple of 5)"; applying De Morgan again to the inner "and" gives "not multiple of 2 or not multiple of 5".

> **Investigate further:** Nested compound statements need De Morgan applied at every level, working from the outside in.

8. True or false: "not(A and (B or C)) is the same as (not A) or ((not B) and (not C))."

Answer: True.

Method: Outer De Morgan: not(A and (B or C)) = (not A) or not(B or C). Inner De Morgan on the second term: not(B or C) = (not B) and (not C). Combining gives exactly the claimed form.

> **Investigate further:** This is the general two-level pattern that B7 applies to a concrete divisibility statement.

9. Negate: "x is rational, or (x is irrational and $x > 0$)."

Answer: "x is irrational, and (x is rational or $x \leq 0$)."

Method: Outer De Morgan: not(P or Q) = (not P) and (not Q), giving "x irrational and not(x irrational and $x > 0$)". Inner De Morgan: not(x irrational and $x{>}0$) = (x rational) or ($x \leq 0$).

> **Investigate further:** The raw De Morgan output is correct but not fully simplified — since "x irrational" and "x rational" cannot both hold, the "x rational" branch is automatically false here, so the negation is equivalent to "x is irrational and $x \leq 0$". Recognising when a mechanically-correct negation simplifies further is a genuine extra step.

10. Negate: "For every prime $p > 2$, p is odd." Is the negation true or false?

Answer: Negation: "There exists a prime $p > 2$ such that p is even." The negation is false.

Method: Negating $\forall p, P(p)$ gives $\exists p, \text{not } P(p)$. The negation is false because the original is true: 2 is the only even prime, and it is explicitly excluded by $p > 2$.

> **Investigate further:** A clean closer: the original statement, the negation, and the underlying fact (2 is the unique even prime) all reinforce each other.

Section C Substitution & Structure

1. For real x and positive integer n, consider: I. For all x and for all n, $x^2 < n$. II. For all x, there exists n such that $x^2 < n$. III. There exists x such that for all n, $x^2 < n$. Which of I, II, III are true?

Answer: II and III only.

Method: I is false: $x = 5, n = 1$ gives $x^2 = 25 \not< 1$. II is true: for any real x, x^2 is a fixed number, so a positive integer $n > x^2$ always exists (take n to be the smallest integer exceeding x^2). III is true: taking $x = 0$, $x^2 = 0 < n$ holds for every positive integer n.

> **Investigate further:** III is the subtle one — the witness $x = 0$ works precisely because n ranges only over *positive* integers (so $n \geq 1$ always); restricting the quantifier's domain is what makes III true here.

2. Determine the complete set of real values of k for which the following is true: "For all real x, if $x < k$, then $x^2 > k$." *(after TMUA 2022 Paper 2 Q9)*

Answer: C) $k \leq 0$.

Method: Case $k \leq 0$: for any $x < k$, $x^2 \geq 0$. If $k < 0$, then $x^2 \geq 0 > k$ automatically. If $k = 0$, then $x < 0$ means $x \neq 0$, so $x^2 > 0 = k$ strictly. Either way the implication holds for every $x < k$. Case $k > 0$: take $x = 0$. Then $x < k$ holds (since $k > 0$), but $x^2 = 0$, and $0 > k$ is false since $k > 0$ — so the implication fails at $x = 0$. Hence the statement is true exactly for $k \leq 0$, option C. Option B ($k < 0$) misses the $k = 0$ boundary; option D ($k \leq 1$) overextends past where the argument still works; option A wrongly assumes no k works at all (the real source question's own answer, for a subtly different claim); option E ignores case $k > 0$ entirely; option F has the case split backwards.

> **Investigate further:** This mirrors the real exam question's own case-split discipline (test $k \leq 0$ and $k > 0$ separately, and produce an explicit witness in each case) — here the two cases genuinely split the answer rather than eliminating every k.

3. Which is the correct negation of "For all primes p, p is odd or $p = 2$"?

Answer: B) There exists a prime p such that p is not odd and $p \neq 2$.

Method: Negating $\forall p, (P(p) \text{ or } Q(p))$ requires two steps: flip the quantifier ($\forall \to \exists$), then apply De Morgan to the inner "or" ($\text{not}(P \text{ or } Q) = (\text{not } P) \text{ and } (\text{not } Q)$). Each wrong option skips exactly one step: A flips the quantifier but leaves the inner clause completely unnegated; C applies De Morgan correctly but leaves the quantifier as $\forall$; D flips the quantifier correctly but keeps "or" instead of switching to "and".

> **Investigate further:** This MCQ is designed so every single-step shortcut lands on a plausible wrong answer — both steps are required together.

4. Negate: "There exists a positive integer n such that $n^2 + n + 1$ is even." Is the negation true or false?

Answer: Negation: "For every positive integer n, $n^2 + n + 1$ is odd." The negation is true.

Method: $n^2 + n + 1 = n(n + 1) + 1$. Since $n(n + 1)$ is a product of two consecutive integers, it is always even, so $n^2 + n + 1$ is always one more than an even number, i.e. always odd. The original existential claim is therefore false, and its negation is true.

> **Investigate further:** This is a concrete instance where checking the underlying mathematics (not just the negation mechanics) is what actually settles the truth value.

5. Negate: "There exists a positive integer n such that for every positive integer $k \leq 10$, k divides n." Is the original statement true or false?

Answer: Negation: "For every positive integer n, there exists a positive integer $k \leq 10$ such that k does not divide n." The original statement is true.

Method: Negating $\exists n\, \forall k{\leq}10, P(n, k)$ gives $\forall n\, \exists k{\leq}10, \text{not } P(n, k)$. The original is true: $n = \mathrm{lcm}(1, 2, \ldots, 10) = 2520$ is divisible by every k from 1 to 10.

> **Investigate further:** Since the original is true, the negation here is correctly false — not every n fails to be divisible by some $k \leq 10$; $n = 2520$ (and all its multiples) is a counterexample to the negation.

6. Consider the statement about a real number x: "There exists a real number y such that $xy - x + y$ is positive." For how many real values of x is this true? *(after TMUA 2022 Paper 2 Q13)*

Answer: E) all values of x.

Method: Rewrite $xy - x + y$ as a function of y: $xy + y - x = y(x+1) - x$, of the form $my + c$ with $m = x + 1$, $c = -x$. If $m \neq 0$ (i.e. $x \neq -1$), this linear function of y takes every real value as y varies, including positive ones, so a suitable y exists. If $m = 0$ (i.e. $x = -1$), the expression is constantly $c = -x = 1$, which is itself positive — so a suitable y (in fact every y) still exists. Every case gives a true statement, option E. The tempting wrong answer is C ("all except exactly one value"): the real exam question this is adapted from has exactly that answer, with its degenerate case failing rather than trivially succeeding — picking C without re-checking the edge case here would be pattern-matching against the wrong source, not against this question's actual arithmetic.

> **Investigate further:** The edge case ($x = -1$, where the "line in y" degenerates to a constant) is exactly the kind of boundary a hasty "it's linear so obviously true" argument would skip — checking it explicitly is what turns a plausible answer into a proven one.

7. Negate: "There exists a real number x such that for every real number y, $xy = y$." Is the original statement true or false?

Answer: Negation: "For every real number x, there exists a real number y such that $xy \neq y$." The original statement is true.

Method: Negating $\exists x\, \forall y, P(x, y)$ gives $\forall x\, \exists y, \text{not } P(x, y)$. The original is true: taking $x = 1$, $1 \cdot y = y$ holds for every real y, so $x = 1$ is a witness.

> **Investigate further:** Recognising $x = 1$ as "the number that does nothing under multiplication" is the same instinct needed later for identity elements in more general algebraic structures.

8. Negate: "For every positive integer n, there exists a positive integer $m < n$ that is prime." Consider the cases $n = 1$ and $n = 2$ carefully. Is the original statement true or false?

Answer: Negation: "There exists a positive integer n such that no positive integer $m < n$ is prime." The original statement is false, witnessed by $n = 1$ (and also $n = 2$).

Method: At $n = 1$, there is no positive integer $m < 1$ at all, so no prime $m < 1$ can exist — the original's claim fails immediately. At $n = 2$, the only candidate is $m = 1$, which is not prime, so the claim fails there too. Both are valid witnesses for the negation.

> **Investigate further:** Small edge cases ($n = 1$ especially) are the most common place a "for every n" claim quietly breaks — a pattern this pillar returns to when hunting counterexamples.

Section D Challenge

1. Consider the statement: "For every finite sequence of positive integers, there exists a positive integer m such that every term of the sequence is at most m." Which of the following is the correct negation?

Answer: A) There exists a finite sequence of positive integers such that, for every positive integer m, some term exceeds m.

Method: Negating $\forall S\, \exists m\, \forall t{\in}S, (t \le m)$ gives $\exists S\, \forall m\, \exists t{\in}S, (t > m)$ — exactly option A. Option B fails to negate "at most m"; option C fails to flip the outer quantifier; option D illegitimately changes "finite" to "infinite", which is not part of a correct negation at all.

> **Investigate further:** The original statement is in fact true (any finite sequence has a maximum term, so m can be taken to be that maximum) — this question tests only the negation construction, not the truth value.

2. Determine, with brief justification, whether the following is true or false: "For every positive real number ε, there exists a positive integer N such that for every integer $n \ge N$, $\dfrac{1}{n} < \varepsilon$."

Answer: True.

Method: Given any $\varepsilon > 0$, choose N to be any positive integer greater than $1/\varepsilon$. Then for every integer $n \ge N$, $\frac{1}{n} \le \frac{1}{N} < \varepsilon$. Such an N always exists because the positive integers are unbounded above (the same Archimedean idea as C1's II).

> **Investigate further:** This is precisely the formal statement that $1/n \to 0$ as $n \to \infty$ — the $\forall\varepsilon\, \exists N\, \forall n$ pattern here is the template for every "limit equals a value" definition met later in analysis.

3. A selection S of n terms is taken from the arithmetic sequence $2, 5, 8, 11, \ldots, 92$. What is the smallest value of n for which S must contain two distinct terms summing to 94? *(after TMUA 2022 Paper 2 Q8)*

Answer: C) $n = 17$.

Method: The sequence has $\frac{92-2}{3} + 1 = 31$ terms. Pair up terms summing to 94: term x pairs with $94 - x$, and both lie in the sequence exactly when $2 \le x \le 92$ (since $94 - x$ then also lies in $[2, 92]$ and is $\equiv 2 \pmod 3$). The self-paired term is $x = 47$ (since $94 - 47 = 47$), which cannot form a distinct pair. Excluding 47, the remaining 30 terms split into 15 pairs: $(2, 92), (5, 89), \ldots, (44, 50)$. A selection avoiding any such pair can include at most one term from each of the 15 pairs, plus the unpaired 47 itself: at most $15 + 1 = 16$ terms. So 17 terms forces a pair, option C. This bound is tight: the 16-term selection $\{47\} \cup \{2, 5, \ldots, 44\}$ (one term per pair, plus 47) contains no two distinct terms summing to 94. Option B (16) is exactly that tight bound, tempting if you forget it's the largest *avoiding* size, not the forcing size. Option A (15) is the pair count alone, forgetting the unpaired 47. Option D (18) overshoots by one, as if the tight bound were 17 rather than 16. Option E (24) has no clean derivation from this argument. Option F (31) is just the total term count, ignoring the pairing structure entirely.

> **Investigate further:** The self-paired "middle" term (47 here, the value with $94 - 47 = 47$) is easy to forget — omitting it from the excluded count would silently overcount how many terms a pair-avoiding selection can hold.

4. Statement $(*)$: "For every finite set of primes $\{p_1, \ldots, p_k\}$, there exists a prime not in the set." (a) Negate $(*)$. (b) Is $(*)$ true or false? Justify briefly.

Answer: (a) "There exists a finite set of primes containing every prime." (b) $(*)$ is true.

Method: (a) Negating $\forall\{p_1, \ldots, p_k\}\, \exists p \notin \{p_1, \ldots, p_k\}$ gives $\exists\{p_1, \ldots, p_k\}\, \forall p, p \in \{p_1, \ldots, p_k\}$, i.e. a finite set containing all primes. (b) $(*)$ is true: given any finite list of primes $p_1, \ldots, p_k$, the number $N = p_1 p_2 \cdots p_k + 1$ is not divisible by any of $p_1, \ldots, p_k$ (each leaves remainder 1), so any prime factor of

N is a prime outside the list, proving one always exists.

> **Investigate further:** $(*)$ and its negation are exactly the infinitude of primes stated two ways — Euclid's construction ($N = p_1 \cdots p_k + 1$) is the whole proof in one line, and it is worth having ready on sight rather than re-derived from scratch.

5. Let $f : \mathbb{R} \to \mathbb{R}$. Consider: I. "For every x, there exists y such that $f(y) = x$." II. "There exists y such that for every x, $f(y) = x$." (a) Are I and II logically equivalent? (b) Give a concrete function f for which I is true but II is false.

Answer: (a) No, not equivalent. (b) $f(y) = y$ (the identity function on $\mathbb{R}$).

Method: (a) I says f is surjective (every x has a preimage, possibly different y for different x); II says a single y produces every possible x, which is impossible for an actual function once more than one value of x is required. (b) For $f(y) = y$: I holds, since for any x, taking $y = x$ gives $f(y) = x$. II fails, since a function assigns each input exactly one output, so no single y can satisfy $f(y) = x$ for every x simultaneously (e.g. $f(y) = 0$ and $f(y) = 1$ cannot both hold).

> **Investigate further:** This closes the day exactly where A9 opened it, now dressed in function notation: swapping $\exists y\, \forall x$ for $\forall x\, \exists y$ is the difference between "every output is achieved, possibly by different inputs" and "one input achieves every output" — the second is almost never possible for a genuine function.